

From Cellular Difference to Population Reinstatement

Oscillatory Scaling, Inhibitory Waves, and Neural Projection in Self Aware Networks

Micah Blumberg

Self Aware Networks Research Institute

selfawarenetworks.com

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Abstract

Remembering a scene requires more than preserving a molecular or synaptic trace. A locally stored relation must become active across a much larger network while its carrier cells, waveform, and spatial scale change. Established neuroscience supplies parts of this transition: learned dendritic and synaptic structure shapes cellular input-output functions; weak inputs can reset oscillatory phase; excitation and inhibition organize traveling population activity; extracellular fields can alter spike timing; and hippocampal pattern completion can accompany neocortical reinstatement. What remains unresolved is how to describe the joined operation without assuming that each receiving population makes a literal copy.

This paper develops a Self Aware Networks model of **transformed relation preservation**. A compatible input engages a learned cellular transform. The resulting timing, phase, waveform, burst, or omission pattern changes selected receivers. Axonal fan-out, recurrent convergence, inhibition, receiver state, and local field effects then re-express the relation across successive neural arrays. A delayed, suppressed, or missing event can be informative when it is defined relative to a maintained expectation. After the operation is explained in physiological terms, the local typed departure is named a Phase Wave Differential (PWD), and the repeated receive-transform-project process is related to Neural Array Projection Oscillation Tomography (NAPOT).

The model separates synaptic, axonal, interneuronal, refractory, and ephaptic routes; distinguishes a signed differential from a negative voltage trace; and reserves *soliton* for patterns that pass shape-preservation, nonlinear-dispersion, perturbation-recovery, and collision tests. A deterministic constructed example tracks a signed relation through two receiver-dependent transforms. Decoding remains high at the third array (0.879), while rate-only (0.612), topology-shuffled (0.658), and negative-channel-ablated (0.583) controls lose substantial information. These are proof-of-concept values, not neural estimates. The biological proposal survives only if a learned relation remains conditionally decodable across identified arrays and causal perturbations selectively disrupt and rescue its predicted route.

Keywords: neural oscillations; traveling waves; memory reinstatement; inhibition; ephaptic coupling; Phase Wave Differential; NAPOT; Self Aware Networks

1. The Scale Problem in an Ordinary Memory

Look at a familiar room, close your eyes, and recall where the door, window, and table are. The original sensory pattern reached many cells. The later memory is not a replay from one preserved synapse. It is a coordinated population event that re-establishes task-relevant relations among features.

This creates a scale problem:

How does a relation stored in local cellular and synaptic organization become recoverable as a

larger active relation without requiring a literal copy at every stage?

The question is not answered by saying that a memory has an engram or that activity broadcasts. Those labels identify substrates and phenomena but do not specify what survives the transition, how negative or omitted events participate, or how one can tell relation preservation from common input, volume conduction, or decoder leakage.

The proposed operation can first be stated without SAN terminology:

1. learning configures a local transform;
2. a compatible input evokes a timed and shaped cellular difference;
3. connected receivers transform that difference according to their own anatomy, phase, gain, and inhibition;
4. excitation and inhibition select a spatial-temporal route;
5. recurrence and convergence extend causal reach; and
6. the same task-relevant relation remains decodable even though its physical carrier changes.

2. The Photocopier and the Black Spot

The 2022 SAN whitepaper used a photocopier to make the scale transition visible. A small pattern placed on the glass can be enlarged through successive copies. A neural array is not a literal copier: each receiver has its own learned morphology, delays, weights, phase, excitability, and inhibitory context. Every stage therefore performs a transformed re-expression.

The black spot adds a second insight. A mark or defect on copier glass appears in the output even when it is not part of the original page. In a neural sequence, the informative feature can similarly be an event that fails to occur when the system strongly expected it. If downstream populations preserve where and when that omission occurred, a negative relation has been projected.

The metaphor earns scientific value only after its variables are measurable. The copy becomes a receiver-dependent operator. The black spot becomes a signed departure from a declared expectation. Enlargement becomes increased spatial reach or downstream causal influence, not necessarily larger raw voltage.

3. Source-Faithful Genealogy

3.1 Local memory and electrical readout

Micah Blumberg's 2012 SAN sources proposed that neuronal morphology functions as a program and that potential difference can provide a sparse readout of the richer dendritic and cellular state. The intended mechanism is functional: learned structure constrains recognition, and a matching input evokes an electrical consequence. The anatomy itself does not travel down the axon.

3.2 Projection across arrays

The 2022 SAN corpus joined that local premise to phase-relative output, axonal fan-out, oscillatory coordination, inhibitory selection, and re-expression at successively larger scales. The mature term **Neural Array Projection Oscillation Tomography** names a repeated receive-transform-project operation. The mature **Phase Wave Differential** term names the local typed difference created at a receiver interface. Neither acronym is backdated to 2012.

3.3 Negative and omitted events

The 2022 note a0287zstar.md proposed that a negative-going, delayed, suppressed, or omitted event relative to tonic activity can carry information. It used the historical phrases *anti-spike*, *anti-matter action potential*, and *negative soliton wave*. The useful operation is a complementary signed pattern. *Anti-matter* is a metaphor and makes no particle-physics claim. *Soliton* remains an unconfirmed high-burden hypothesis.

3.4 Burst-to-memory scaling

The 2022 a0309z.md note connected bursts, sequential spiking, inhibitory consequences, memory reactivation, and larger pattern recruitment. Where the source compressed this relationship as spikes inhibiting neighbors, the recovered physiological chain is:

source spike or burst

- > excitatory transmission and collateral recruitment
- > interneuronal, refractory, synaptic, or ephaptic inhibition
- > selected receivers are suppressed or retimed
- > excitation and inhibition organize a moving contrast
- > later events are allocated within the altered route

This intent-recovery step preserves the historical proposition while separating distinct biological mechanisms.

4. Established Components of the Bridge

4.1 Learned cellular transforms

Active dendrites, nonlinear branch integration, structural plasticity, receptor state, intrinsic excitability, and recent activity all shape cellular response. The companion paper *From Morphology to Phase* owns the detailed local mechanism. This paper begins with its interface: a learned cellular state recognizes an input relation and emits a typed difference.

4.2 Finite phase effects and oscillatory coordination

An input can advance or delay the phase of an oscillator depending on receiver state. Coupled oscillators can lock, drift, or form structured relative phases. The relevant point is not that all memory requires global synchrony. It is that a small, correctly timed event makes a finite change to a larger timing relation, and repeated convergence can preserve or amplify that relation.

4.3 Excitation, inhibition, and traveling organization

Traveling neural activity can coordinate the order and direction of interactions across cortex. Inhibitory circuitry changes the speed, width, stability, and recruitment of such activity. Human recordings show that traveling-wave direction differs between memory encoding and recall, while recent work distinguishes planar, spiral, and concentric wave organizations across memory states. These findings establish population-scale spatiotemporal organization; they do not by themselves identify the SAN relation or prove NAPOT.

4.4 Pattern completion and reinstatement

Hippocampal pattern completion and neocortical reinstatement provide an empirical example of a cue-related relation becoming active beyond its initial carrier. The scientific opportunity is to connect cellular and array-scale measurements: determine whether a relation that is present in a source ensemble can be followed through intermediate transforms into reinstated neocortical content.

4.5 Local electric-field effects

Transmembrane currents contribute to extracellular potential. Under appropriate geometry, endogenous fields can change membrane voltage and spike timing. Rapid ephaptic inhibition has been demonstrated in specialized axon-terminal geometry, and cell classes can differ in their susceptibility to applied fields. Ephaptic coupling is therefore a plausible local term, not a universal replacement for synaptic or axonal transmission.

5. Formal Model: Local Match to Typed Event

Let $\mathbf{s}_i$ describe the learned dendritic and synaptic state of cell i , $\mathbf{u}_i(t)$ its input, and $\mathbf{h}_i(t)$ its recent state. The match is

$$q_i(t) = \mathcal{M}_i(\mathbf{u}_i(t); \mathbf{s}_i, \mathbf{h}_i(t)). \quad (1)$$

The cell's phase follows a receiver-sensitive, delayed coupling model,

$$\dot{\theta}_i = \omega_i + \sum_j K_{ij} H_{ij}(\theta_j - \theta_i - \tau_{ij}) + Z_i(\theta_i) I_i(t) + \eta_i(t). \quad (2)$$

The output is a typed event vector,

$$\mathbf{y}_i(t) = \mathcal{G}_i(q_i, \theta_i, \dot{\theta}_i, V_i, T_{AP,i}, B_i, \mathbf{h}_i), \quad (3)$$

where the declared variables may include phase, latency, action-potential duration, waveform, burst state, and release consequences. Each component must be tied to a measurement; the equation is not permission to treat an unmeasured latent variable as evidence.

6. Receiver-Dependent Projection

For source array A and receiving array B ,

$$\mathbf{x}_B(t + \Delta) = \mathcal{P}_{A \rightarrow B}(\mathbf{y}_A(t), W_{AB}, \tau_{AB}, \mathbf{g}_B(t), \mathbf{I}_B(t), \phi_e(t)). \quad (4)$$

The operator $\mathcal{P}_{A \rightarrow B}$ is not a copier. It includes anatomical connectivity W_{AB} , delays τ_{AB} , receiver gain $\mathbf{g}_B$, inhibitory state $\mathbf{I}_B$, and local extracellular potential ϕ_e . The same source pattern can therefore produce different receiver patterns while preserving a task-relevant relation.

Let $\hat{x}_{i,T}(t)$ be a preregistered expectation under tonic state T . Define

$$d_i(t) = x_i(t) - \hat{x}_{i,T}(t), \quad p_i(t) = \max(0, d_i), \quad n_i(t) = \max(0, -d_i). \quad (5)$$

Only after this reference-dependent operation is explicit do we call the typed local departure a **Phase Wave Differential**. A negative channel can represent suppression, delay, or omission, but the sign must be interpreted relative to the measured variable, electrode geometry, and baseline. A negative extracellular deflection is not automatically inhibition.

Repeated receiver-dependent projection across identified neural arrays is the operational core associated with **NAPOT**. Its empirical signature is transformed relation preservation, not visual resemblance between raw traces.

7. Excitatory-Inhibitory and Ephaptic Terms

A standard neural-field comparator is

$$\tau_E \partial_t E = -E + F(w_{EE} * E - w_{EI} * I + S), \quad (6)$$

$$\tau_I \partial_t I = -I + G(w_{IE} * E - w_{II} * I). \quad (7)$$

These equations can produce traveling excitatory and inhibitory patterns without invoking SAN. The SAN test asks whether a learned signed relation survives across identified receive-transform-project stages beyond what this generic field model, common input, or rate alone predicts.

Extracellular potential can be represented as the superposition of transmembrane currents through conductive geometry,

$$\phi_e(\mathbf{r}, t) = \sum_i \int G_i(\mathbf{r}, \mathbf{r}') I_{m,i}(\mathbf{r}', t) d\mathbf{r}'. \quad (8)$$

A morphology-relative ephaptic influence on cell i is

$$I_{\text{eph},i}(t) = \gamma_i [\phi_e(\mathbf{r}_{i,a}, t) - \phi_e(\mathbf{r}_{i,b}, t)]. \quad (9)$$

The voltage difference avoids treating absolute extracellular potential as a reference-free causal quantity. Synaptic, axonal, gap-junction, refractory, neuromodulatory, and ephaptic contributions must be perturbed separately wherever possible.

8. Measuring Relation Preservation and Scaling

Let z be a task relation that can be defined independently of the exact carrier cells. At array k , decoder D_k estimates that relation. Conditional preservation is

$$\mathcal{R}_{k \rightarrow k+1} = I(D_k(\mathbf{x}_k); D_{k+1}(\mathbf{x}_{k+1}) \mid \mathbf{c}), \quad (10)$$

where $\mathbf{c}$ includes stimulus identity, movement, arousal, trial time, firing rate, and common-input controls. Cross-validated representational similarity or cross-decoding can replace conditional mutual information if selected before analysis.

Scaling gain concerns reach or causal influence rather than raw amplitude:

$$G_{A \rightarrow B} = \frac{\Delta I(z; \mathbf{x}_B | \mathbf{c})}{\max(\varepsilon, \Delta I(z; \mathbf{y}_A | \mathbf{c}))}. \quad (11)$$

Here $\varepsilon > 0$ is a preregistered numerical floor, and the absolute numerator and denominator must always be reported beside the ratio. A value above one does not create information from nothing. It can arise because a weak source perturbation gates a larger receiver population whose state already contains relevant structure. The numerator and denominator must therefore be interpreted as intervention-dependent changes in decodable task information, not as a violation of data processing. If the source change is indistinguishable from zero, the ratio is reported as undefined rather than as amplification.

9. When a Negative Pattern Is a Wave or Soliton

For the negative field $n(\mathbf{r}, t)$, fit a velocity $\mathbf{v}$ and amplitude factor a :

$$\epsilon_{\text{wave}} = \frac{\|n(\mathbf{r}, t + \Delta t) - a n(\mathbf{r} - \mathbf{v}\Delta t, t)\|_2}{\|n(\mathbf{r}, t)\|_2}. \quad (12)$$

A traveling inhibitory wave requires low held-out translation error, coherent direction and speed, and causal effects at predicted receivers. A **negative soliton** carries a higher burden. It must show bounded shape error, a nonlinear amplitude-speed or amplitude-width relation, resistance to ordinary dispersive broadening, recovery after bounded perturbation, and characteristic collision behavior. The governing nonlinear equation must be selected from measured dispersion and nonlinearity rather than imposed because the historical word *soliton* appears in a source note.

10. Deterministic Cross-Array Demonstration

10.1 Design

The accompanying model constructs 6,000 trials containing a 12-position signed relation. Only below-tonic positions carry the class-defining relation; matched positive activity is intentionally nondiagnostic. Array A emits the source relation. Array B applies a receiver-dependent mixing and inhibitory transform. Array C applies a second transform and noise. Decoders are evaluated on held-out data under fixed seed 20260808.

This fixture is deliberately stronger than the first attempted design. In the first design, removing the negative channel improved classification, proving that the proposed ablation was not causal in that construction. That fixture was rejected. The retained design requires the negative channel for relation recovery.

10.2 Results

Signed decoding was 1.000 at array A , 0.967 at array B , and 0.879 at array C . At array C , rate-only decoding fell to 0.612; shuffling the topological relation produced 0.658; ablating the negative channel produced 0.583. The signed relation therefore survived two transforms better than three declared controls.

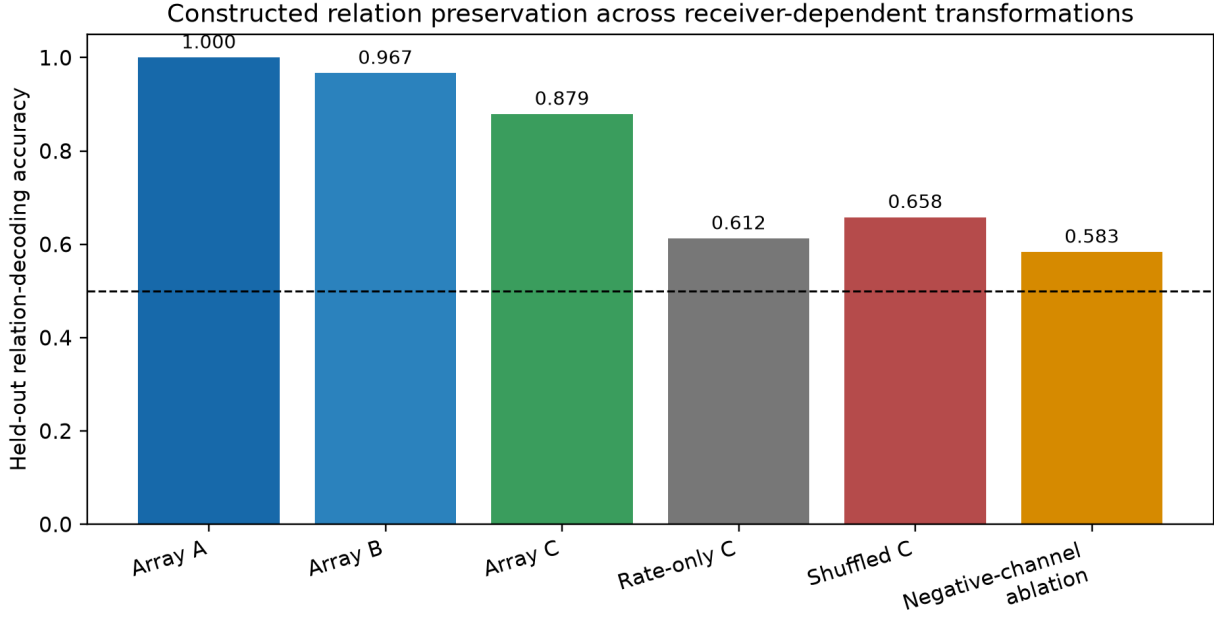


Figure 1: Constructed cross-array signed-relation demonstration

The model does not establish a biological traveling wave, an ephaptic route, a soliton, an engram, or consciousness. It establishes only that the proposed relation-preservation discriminator can be made executable and can reject rate-only, topology-shuffled, and negative-channel-ablated alternatives in a transparent construction.

All declared checks pass: the signed relation survives two transforms, signed decoding exceeds rate-only decoding, stable topology matters, and the negative channel is causally relevant in the retained fixture. The machine-readable receipt records the model class, seed, metrics, checks, artifact paths, and source-script SHA-256.

11. Experimental Ladder

11.1 Dendritic match to cellular event

Identify a learned cue relation at dendritic branches. Measure branch voltage or calcium, somatic voltage, spike timing, bursts, and waveform duration. Test whether a morphology-sensitive match predicts typed output beyond total input count and mean rate.

11.2 Cellular event to local inhibitory consequence

Perturb a source spike or burst at controlled phases. Record identified excitatory cells, inhibitory interneuron classes, and extracellular voltage. Decompose synaptic, refractory, interneuronal, and ephaptic timing effects instead of calling every negative consequence inhibition.

11.3 Signed pattern across space

Estimate the tonic expectation at every site, then map positive and negative differentials with dense spatial sampling. Test direction, speed, held-out shape error, common-input controls, and receiver perturbation.

11.4 Sharp-wave-ripple component test

Resolve sharp-wave current, ripple-band activity, pyramidal spikes, and interneuron spikes. Decode the learned relation from each component and from downstream targets. Manipulate interneuron subclasses to test spike allocation and relation preservation. A sharp-wave ripple is an empirical test bed, not a synonym for NAPOT.

11.5 Array-to-array transformed reinstatement

Track at least three anatomically connected arrays. Train scale-specific and cross-scale decoders on a relation independent of exact cell identity. Compare the receiver-dependent model with rate-only, topology-shuffled, common-input, and direct-broadcast alternatives.

11.6 Black-spot omission test

Establish a predictable sequence, omit one event without changing surrounding positive events, and test whether the omission remains decodable downstream. Perturb the route predicted to carry the omission and attempt matched timing rescue.

11.7 Ephaptic route test

Measure local field geometry and morphology-relative voltage differences. Cancel, reverse, or phase-shift the field while preserving synaptic stimulation as closely as possible. Test whether predicted receiver timing changes disappear and can be rescued.

11.8 Soliton eligibility test

Measure the candidate pattern at sufficient spatial and temporal resolution. Vary amplitude, width, medium state, and inhibitory gain. Compare dispersion, recovery, and collision behavior against passive filtering, reaction-diffusion, neural-field, and sequential common-input models.

12. Controls and Rival Explanations

The principal alternatives are:

- ordinary axonal broadcast without relation-preserving transformation;
- pattern completion confined to one recurrent circuit;
- common input producing similarity at multiple sites;
- volume conduction or reference mixing producing an apparent wave;
- passive filtering producing a negative-going trace;
- refractory dynamics without recruited inhibition;
- generic inhibition-controlled traveling activity without a memory-specific signed relation; and
- decoder leakage from movement, respiration, arousal, stimulus identity, or trial time.

Every positive result requires topology shuffles, temporal jitter, matched-rate controls, cell-class and layer controls, reference-sensitivity checks, and held-out perturbation. A compelling result must show not just decodability but route-specific causal loss and rescue.

12.1 Preregistered evidence ladder

Level	Required observation	Warranted conclusion
A	relation decodes in one array	local representation is present
B	relation decodes across identified arrays after controls	transformed preservation is present
C	route-specific perturbation selectively reduces preservation	the route contributes causally
D	matched timing or field rescue restores preservation	the specified route model is supported
E	propagation and nonlinear tests pass	wave or soliton language is warranted at the tested level

The full NAPOT-scale claim requires evidence from several levels; no single oscillatory correlation licenses the entire chain.

13. Falsification Conditions

The joined scale claim is weakened or rejected if the task relation cannot be decoded after the first projection once common input, firing rate, anatomy, and trial timing are controlled. The signed-code claim fails if omissions or suppressions add no information beyond positive activity and baseline state. The ephaptic term fails where a field contribution is predicted but field cancellation produces no receiver-timing effect. The wave claim fails if a static or common-input model explains the pattern. The soliton claim fails if the pattern disperses normally and lacks nonlinear recovery or interaction.

One failed route does not automatically erase every other route. Conversely, evidence for one route does not prove the complete NAPOT architecture. The paper is a joined mechanism with separable experimental burdens.

14. Discussion

The key shift is from copying activity to preserving relations through transformation. A memory can survive even when its exact cells and waveform change, provided that the receiver can recover the same task-relevant constraint. This is analogous to translating a sentence into another language: the symbols change while an operation or relation is preserved. In neural tissue, the translation is physical and receiver-dependent.

Signed departures are central to the proposal. Meaning is not restricted to extra spikes. A well-predicted event that fails to occur can change downstream timing, release inhibition, or redirect a sequence. That insight connects the historical black-spot metaphor to a concrete omission experiment.

The model also prevents a common amplitude mistake. Scaling does not require a small cellular voltage to become a proportionally larger field by passive addition. A precisely timed cellular event can gate a receiver population whose existing energy, structure, and recurrent gain generate a larger causal consequence. The source contributes information and timing; the receiving system supplies most of the energy.

15. Conclusion

A local memory becomes useful only when its relation can be made active elsewhere. The proposed SAN bridge is a sequence of learned match, typed difference, receiver-dependent projection, signed route selection, recurrence, and transformed reinstatement. It is stronger than a metaphor because each stage has a measurable variable, comparator, perturbation, and failure condition. If a learned relation survives multiple identified transforms and selective disruption and rescue track the predicted route, the scale bridge will have empirical support. If it does not, the model must be revised at the failed stage.

Research Ethics and Data Scope

This theoretical paper reports no new human or animal experiment. Future invasive recordings and perturbations require appropriate institutional review, welfare protections, preregistered stopping and exclusion rules, and explicit separation of clinical electrode placement from research aims. The numerical results in this manuscript come from constructed data generated by the included deterministic script.

Code, Data, and Reproducibility

The deterministic demonstration, fixed seed, machine-readable results, and figure are included with this manuscript:

- `model/run_cross_array_relation_demo.py`
- `model/results/cross_array_relation_results.json`
- `model/results/cross_array_relation_metrics.csv`
- `output/figures/cross-array-signed-relation-demo.png`

The demonstration uses constructed data and makes no biological effect-size claim.

Authorial Provenance and AI Assistance

The morphology-to-readout ancestry is preserved in Micah Blumberg’s 2012 source record. The PWD, NAPOT, photocopier, anti-spike, inhibitory-wave, and burst-to-scaling formulations are preserved in the 2022 SAN corpus. AI tools assisted source recovery, mathematical specification, model implementation, and editorial review. The historical theory program and authorship belong to Micah Blumberg.

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